

MODULE 4: LEARNING

Lecture 8: Classical Conditioning: Principles, Processes, and Applications

Types of Animal Behavior and Learning

Animals, including humans, respond to their environments through three broad categories of behavior: reflexes, instincts, and learned behaviors.

1. Reflexes

Reflexes are simple, automatic, and involuntary responses to specific stimuli. They occur without conscious thought and are usually controlled by the spinal cord or lower brain centers.

Key Features:

- Innate (present at birth)
- Very fast and automatic
- Stereotyped (same response every time)
- Protective in nature

Examples:

- Pulling your hand away from a hot stove (withdrawal reflex)
- Blinking when something approaches your eyes
- Knee-jerk response during a medical test

Psychological Importance:

- Help in immediate survival (avoid injury)
- Require no learning or decision-making
- Form the most basic level of behavior

2. Instincts

Instincts are inborn patterns of behavior that are more complex than reflexes and are triggered by certain stimuli. Also called fixed action patterns, they are often goal-directed and involve sequences of actions.

Key Features:

- Innate but more complex than reflexes
- Universal within a species
- Triggered by specific stimuli (called releasers)

- Often involve fixed action patterns

Examples:

- Infants sucking for nourishment
- Birds building nests
- Humans showing fight-or-flight response
- Animals migrating seasonally

Psychological Importance:

- Aid in survival and reproduction
- Provide pre-programmed behaviors
- Bridge the gap between reflexes and learning

3. Learned Behaviors

Psychologists define learning as a relatively permanent change in a person's knowledge or behavior that results from experience. It implies a relatively permanent change in behavior (or capacity for behavior) due to experience.

This definition contains three critical elements that distinguish learning from other types of changes:

- It creates lasting change, not something that disappears after minutes or hours;
- What changes is either the knowledge stored in memory or actual behavior; and
- The source of change is experience and interaction with the environment, not temporary states like fatigue, motivation, drugs, or physical condition.

Learning vs. Maturation

Maturation refers to the natural, biological growth processes that occur in all living organisms over time. Maturation is not influenced by experience or practice but is driven by genetic factors.

The key difference between learning and maturation is that learning is an acquired skill, while maturation is a natural progression. Learning builds upon maturation, but it's not entirely dependent on it.

You can have matured in certain areas, like physical development, but still need to learn other aspects, like social skills or cognitive skills. Learning and maturation are not complete opposites — they work together. When a person's biological components (brain/body) are properly developed at certain stages, learning can be promoted.

Example: Language Acquisition

At a certain age, we begin to talk and speak a language because our brains have matured to a level where they can recognize and imitate human speech — here, maturation creates the readiness, and then learning takes over.

Types of Learning

Learning can be classified into three main types: associative, nonassociative, and observational. In many situations, multiple forms of learning may occur at the same time.

Associative Learning

Associative learning occurs when an individual forms a connection between two stimuli or between a behavior and its consequence. Two important types of associative learning are classical conditioning and operant conditioning.

- In classical conditioning, we associate pairs of stimuli that occur sequentially in time.
- In operant conditioning, we associate behaviors and their consequences.

This lecture will elaborate on classical conditioning and the next lecture (Lecture 9) will elaborate on operant conditioning.

Nonassociative Learning

Nonassociative learning involves a change in the magnitude of response to a single stimulus, without forming associations with other stimuli. It has two subtypes:

(a) Habituation:

Decreased response to a repeated, harmless stimulus. Example: Ignoring the sound of a fan after hearing it continuously.

(b) Sensitization:

Increased response to a strong or repeated stimulus. Example: Becoming more sensitive to noise after hearing a loud explosion such as following an earthquake, people experience exaggerated responses to movement, light, or noise.

Why would we show habituation to some stimuli and sensitization to others?

In general, habituation occurs in response to milder stimuli, whereas sensitization occurs in response to stronger stimuli.

Habituation ensures that we do not waste precious resources monitoring low-priority stimuli. On the other hand, sensitization is useful in dangerous situations. After detecting one harmful stimulus, raising our overall level of responsiveness should improve reaction time should other dangers arise.

Observational Learning

Observational learning, also known as social learning or modeling, occurs when one organism learns by watching the actions of another organism. This type of learning will be discussed in Lecture 10.

Examples:

- A child learns manners by watching parents
- Students imitate teachers' behavior
- Learning skills by watching videos

Classical Conditioning

Pavlov's Theory of Classical Conditioning

Although learning theory primarily developed in the United States, it was significantly influenced by Russian physiologist Ivan Pavlov. In the early 20th century, Pavlov studied the digestive systems of humans and other animals and observed the salivary responses of dogs.

He noticed that dogs would automatically salivate when given food, a naturally occurring response he termed the unconditioned response, triggered by the unconditioned stimulus, which is the food.

Pavlov discovered that if a neutral stimulus, such as a light or bell, preceded the food, the dogs would eventually salivate in response to the light or bell alone (called a conditioned response).

This process, where a neutral stimulus becomes associated with an unconditioned stimulus to elicit a conditioned response, is known as classical conditioning. Pavlov's extensive research in this area laid the foundation for classical conditioning, a core concept in learning theory

1 Natural response (before conditioning occurs)

Stimulus	Response
Food presented to dog [Food = Unconditioned stimulus]	Dog salivates [Salivation = Unconditioned response]
Light is switched on [Light = Neutral stimulus]	No response from the dog

2 Conditioning procedure for several trials

Stimulus	Response
Light is switched on and food is presented to dog Neutral stimulus + Unconditioned stimulus	Dog salivates Unconditioned response

3 After the conditioning trials

Stimulus	Response
Light alone is presented to dog Conditioned stimulus	Dog salivates Conditioned response

(Source: Maltby, Day & Macaskill (2017))

Examples of Classical Conditioning in Human Life

1. Fear Response

A person who has experienced a traumatic event, such as being in a car accident, might develop a fear response when hearing the sound of screeching tires. Initially, the sound of screeching tires (neutral stimulus) does not elicit fear. However, during the accident, the sound of screeching tires (now an

unconditioned stimulus) is paired with the fear and anxiety of the accident (unconditioned response). After conditioning, the sound of screeching tires becomes a conditioned stimulus, eliciting a fear response (conditioned response) even when the person hears it in a non-threatening context.

2. Taste Aversion

Take an example of someone who gets food poisoning after eating sushi. Before the incident, sushi is a neutral stimulus. When the person eats sushi and then experiences nausea and vomiting due to food poisoning (unconditioned stimulus), the resulting illness is the unconditioned response. After this event, the person might feel nauseous (conditioned response) at the mere sight or smell of sushi (conditioned stimulus), even if the sushi is perfectly safe to eat. This learned aversion can extend to other foods with similar textures or smells, demonstrating generalization.

Key Processes in Classical Conditioning

1. Generalization

Definition: Generalization occurs when a conditioned response (CR) is elicited by stimuli that are similar to the conditioned stimulus (CS), even if those stimuli were never paired with the unconditioned stimulus (US).

Example: If a dog is conditioned to salivate to the sound of a bell (CS), it might also salivate to similar sounds, like a chime or a different type of bell, even if those sounds were never directly associated with food (US).

2. Discrimination

Definition: Discrimination is the ability to differentiate between the conditioned stimulus (CS) and other stimuli that have not been paired with the unconditioned stimulus (US). This means that the conditioned response (CR) occurs only in response to the specific conditioned stimulus (CS) and not to similar stimuli.

Example: If a dog is conditioned to salivate to the sound of a specific bell (CS), it learns to salivate only to that specific bell and not to other similar sounds, like a chime or different bell tones, if those sounds are not paired with food (US).

3. Extinction

Definition: Extinction occurs when the conditioned stimulus (CS) is repeatedly presented without the unconditioned stimulus (US), leading to a gradual weakening and eventual disappearance of the conditioned response (CR).

Example: If the sound of the bell (CS) is repeatedly presented to the dog without following it with food (US), the dog will eventually stop salivating (CR) in response to the bell.

4. Spontaneous Recovery

Definition: Spontaneous recovery is the reappearance of a previously extinguished conditioned response (CR) after a period of rest and no further conditioning. This shows that the learned association is not entirely erased but can re-emerge after some time.

Example: After the conditioned response (salivation) to the bell (CS) has been extinguished, if some time passes and the bell is presented again, the dog may start to salivate (CR) again, even if the food (US) is still not presented.

Pavlov demonstrated that a conditioned response could generalize to similar stimuli. For instance, dogs might respond to changes in light as they do to the original stimulus. In the context of children, reading on a parent's lap could generalize to reading across a room and eventually to reading independently and enjoying it.

However, Pavlov also showed limits to generalization; dogs learned to discriminate between sounds associated with food and those that were not. This discrimination meant they would only salivate to specific 'food' sounds.

Additionally, Pavlov demonstrated that conditioning could be reversed. If the conditioned stimulus (like a light) is repeatedly presented without the unconditioned stimulus (food), the conditioned response (salivation) diminishes and eventually ceases, a process known as extinction.

Limitations of Classical Conditioning

Modern research indicates that classical conditioning is not as straightforward as Pavlov initially assumed. While Pavlov believed that conditioning principles were general rules applicable to all animals, it's now understood that different organisms are more easily conditioned to respond in particular ways to specific stimuli (Garcia & Koelling, 1966).

For instance, hungry dogs can be conditioned to salivate with a bell that is paired with the sight and smell of meat. However, each species and even each individual within a species has certain tendencies that either facilitate or impede certain types of learning. For example, humans tend to rely more on visual cues than smells, and different people have varied perceptual and aesthetic inclinations.

Despite these complexities, classical conditioning remains a potent explanation for response patterns, particularly when there is a strong natural pairing of stimulus and automatic response.

John B. Watson and Behaviorism

John B. Watson is widely recognized as the "father of behaviorism." John B. Watson, an American psychologist, was greatly influenced by Pavlov's early work and applied Pavlov's observational techniques in his own research, replicating some of Pavlov's findings in the United States.

Watson advocated for a shift in American psychology toward becoming a true science by rejecting the introspective methods used by Freud and other psychoanalysts, which he deemed unscientific. Watson argued for a focus on observable behavior and the effects of stimuli, using rigorous scientific methods, primarily in laboratory settings, to collect data.

He published his views in the 1914 book "Behaviour" and further elaborated in "Psychology from the Standpoint of a Behaviourist" in 1919, which introduced Pavlov's work to a broader audience. Watson is credited with founding the School of Behaviourism, although his career in psychology ended in 1920

when he left academia for business. His work, along with the popularization of Pavlov's research, paved the way for further developments in behaviorism.

The Little Albert Experiment

Watson is perhaps best known for the "Little Albert" experiment, conducted with his graduate student Rosalie Rayner in 1920. In this experiment, an 11-month-old child named Albert was conditioned to fear a white rat. Initially, Albert showed no fear of the rat (neutral stimulus).

However, Watson and Rayner paired the presentation of the rat with a loud, frightening noise (unconditioned stimulus), which naturally caused Albert to cry (unconditioned response). After several pairings, Albert began to cry (conditioned response) at the sight of the rat (conditioned stimulus), demonstrating that fear could be conditioned in humans. This experiment also showed generalization, as Albert began to show fear responses to other white, furry objects.

Applications in the Real World: Behavior Therapy

For many psychologists, classical conditioning of emotional reactions is crucial in understanding the development of psychopathology and in devising behavioral change strategies. Behavior therapy based on classical conditioning focuses on extinguishing problematic responses, such as conditioned fears, or conditioning new, more desirable responses to stimuli that elicit unwanted reactions like anxiety.

Mary Cover Jones (1924) conducted one of the first systematic behavior therapy experiments, aiming to remove a fear response in a boy named Peter, who was two years and ten months old. Peter, described as generally healthy and well-adjusted, had an exaggerated fear of a white rat, which extended to other white and furry objects like a rabbit, fur coat, feather, and cotton wool. This fear was similar to the fear response conditioned in the Little Albert experiment by John B. Watson and Rosalie Rayner.

Jones meticulously documented Peter's fear responses and the conditions that elicited the greatest fear. She then aimed to "uncondition" Peter's fear response to the rabbit and observe if this unconditioning would generalize to other stimuli. Jones brought Peter to play while the rabbit was present, alongside other children who were not afraid of the rabbit. Gradually, Peter's fear diminished, and he moved from almost complete terror at the sight of the rabbit to a completely positive response.

Mary Cover Jones observed that after successfully unconditioning Peter's fear of the rabbit, he also lost his fear of the fur coat, feathers, and cotton wool. This generalization of the unconditioned response to other stimuli indicated that the procedure was effective even without understanding the origins of Peter's fears. This study marked a significant early use of behavior therapy to treat conditioned fears, demonstrating the potential for classical conditioning principles to facilitate behavioral change.

Systematic Desensitization

A significant advancement in applying classical conditioning principles to psychopathology was the development of systematic desensitization by Joseph Wolpe, a South African psychiatrist influenced by Pavlov's work. Wolpe considered persistent anxiety reactions as learned responses that could be unlearned through therapy.

Systematic desensitization, Wolpe's therapeutic technique, aims to inhibit anxiety through counterconditioning. In counterconditioning, an individual learns a new response that is physiologically incompatible with the existing anxiety or fear response. The goal is for the person to learn a new response, such as relaxation, to the previously feared stimulus.

Through new classical conditioning experiences, the individual is taught to experience relaxation instead of fear when exposed to the stimulus. Once this new response is established, the individual's fear is expected to be eliminated. This technique has become a cornerstone in behavioral therapy for treating anxiety and phobias.

Unconscious Development of Fears and Attitudes

Research has demonstrated that classical conditioning can be used to show how people unconsciously develop fears and attitudes toward others. Studies by Krosnick, Betz, Jussim, Lynn, and Kirschenbaum (1992), and Ohman and Soares (1993) have shown that subliminal presentation of stimuli with positive or negative affective value can influence attitudes toward other stimuli. For example, if a photo is subliminally associated with a negative emotion, people will unconsciously develop a dislike for that photo, and vice versa for positive associations.

Implications for Attitudes and Prejudices

This line of research suggests that many of our attitudes and preferences may be classically conditioned on a subliminal or unconscious basis. Social psychologist Cacioppo (1999) concluded that aversive prejudices, once created, may be difficult to consciously eliminate. People can have egalitarian beliefs but still display prejudicial behaviors in certain situations due to conditioned aversive reactions learned early in childhood. This implies that non-prejudicial attitudes can coexist with conditioned negative reactions, explaining why people may act prejudicially despite sincerely held egalitarian views.

Lecture 9: Operant Conditioning: Principles, Processes, and Applications

Recap: Lecture 8

The previous lecture introduced the core mechanics of learning and classical conditioning:

- Types of Animal Behavior and Learning (Reflexes, Instincts, and Learning)
- Types of learning
- Pavlov's theory of classical conditioning
- Key processes in classical conditioning (Generalization, Discrimination, Extinction, Spontaneous Recovery)
- John B. Watson and behaviorism
- Applications in real world: Behavior therapy and unconscious fear and attitudes

Key Concepts

The Radical Behaviorism of Burrhus Frederic Skinner

Although John Watson left the field of psychology, behaviorism was carried forward by significant figures such as Clark Hull, who developed a systematic drive theory of learning, and John Dollard and Neal Miller, who applied Hull's theory to phenomena involving drives and intrapsychic conflicts, relevant to psychoanalysis.

However, the most influential figure in 20th-century psychology was B.F. Skinner, a Harvard psychologist known for his impact on the field. Skinner, regarded as the most eminent psychologist of the last century, expanded behaviorism beyond learning psychology to a comprehensive philosophy. He articulated broad behavioral principles and developed technologies aimed at improving human experiences, making behaviorism a far-reaching approach.

Skinner was influenced by the research of Pavlov and Watson, among others, and expanded upon their work. While he did not deny the existence of unconscious processes and inner states, he believed it was unscientific and unnecessary to rely on these unobservable processes to explain behavior.

Skinner accepted the principles of classical conditioning but believed they applied to a limited range of learning situations. He argued that the consequences of behavior are crucial to learning, affecting the likelihood of a behavior being repeated. According to Skinner, the classical conditioning model of stimulus-response is too simplistic. He demonstrated that what happens after a response—such as praise or criticism—influences whether the behavior is repeated. For instance, praise for a seminar presentation increases the likelihood of future presentations, while criticism decreases it.

Operant Conditioning & Operants

In Skinner's model, operants refer to behaviors that are influenced by the consequences that follow them. Unlike reflexive responses in classical conditioning, operants are voluntary behaviors that an

organism performs, which then produce effects in the environment. Operants are actions that operate on the environment to produce outcomes, and these outcomes shape future behavior.

Skinner's view is that stimuli in the environment do not force the organism to behave or incite it to act. The initial cause of behavior is in the organism itself.

Skinner's Perspective on Emitted Behavior

"There is no environmental eliciting stimulus for operant behavior; it simply occurs. In the terminology of operant conditioning, operants are emitted by the organism. The dog walks, runs, and romps; the bird flies; the monkey swings from tree to tree; the human infant babbles vocally. In each case, the behavior occurs without any specific eliciting stimulus. It is in the biological nature of organisms to emit operant behavior" (Reynolds, 1968, p. 8).

The Skinner Box

Skinner developed a laboratory apparatus known as the Skinner box to study the effects of reinforcers on behavior. The Skinner box is customized depending on the organism being studied. For instance, a Skinner box designed for rats typically includes a lever that the rat can press and a mechanism for delivering a reinforcer like a food pellet. By presenting the reinforcer, researchers can observe whether it influences the frequency of the behavior, such as lever pressing. Skinner viewed this controlled environment as the ideal setting to study the fundamental principles of behavior.

Process of Operant Conditioning

Reinforcement

In Skinner's model of learning, reinforcement is a central concept. It refers to any consequence that strengthens or increases the likelihood of a behavior being repeated. Reinforcement can be positive or negative, and it can be delivered according to different schedules.

Positive Reinforcement:

This involves presenting a pleasant or rewarding stimulus after a behavior, which increases the likelihood of the behavior being repeated. For example: A teacher praises a student for completing their homework on time. The praise (positive stimulus) reinforces the behavior of timely homework submission.

Negative Reinforcement:

This involves removing an unpleasant stimulus after a behavior, which also increases the likelihood of the behavior being repeated.

- A driver starts regularly obeying the speed limit to avoid receiving traffic tickets (removal of the unpleasant stimulus of getting a ticket).
- A person takes painkillers to relieve a headache. The removal of the headache (unpleasant stimulus) reinforces the behavior of taking the medication.

Punishment

In Skinner's framework of operant conditioning, punishment refers to the process by which a consequence is applied to decrease the likelihood of a behavior recurring in the future. Punishment involves the presentation of an aversive stimulus (positive punishment) or the removal of a pleasant stimulus (negative punishment), both with the intention of reducing the frequency of a particular behavior.

Skinnerians generally are against the use of punishment, whose effects tend to be temporary and whose administration may lead people to rebel against their use. Throughout his career, Skinner emphasized the value of positive reinforcement in shaping behavior.

Schedules of Reinforcement

Reinforcement can be provided according to different schedules, which affect how quickly and how strongly a behavior is learned and maintained.

Continuous Reinforcement:

In this schedule, a behavior is reinforced every time it occurs. This is useful for establishing a new behavior but can lead to rapid extinction if the reinforcement stops.

Intermittent (or Partial) Reinforcement:

In this schedule, a behavior is not reinforced every time it occurs. There are several types of intermittent reinforcement schedules:

- Fixed-Ratio (FR) Schedule: Reinforcement is provided after a fixed number of responses. For example, a salesperson earns a bonus after every 5 sales.
- Variable-Ratio (VR) Schedule: Reinforcement is provided after a variable number of responses, based on an average. For example, a slot machine pays out after an unpredictable number of lever pulls.
- Fixed-Interval (FI) Schedule: Reinforcement is provided for the first response after a fixed period of time has elapsed. For example, a worker receives a paycheck every two weeks.
- Variable-Interval (VI) Schedule: Reinforcement is provided for the first response after varying intervals of time, based on an average. For example, tests or quizzes occur at unpredictable time intervals in the class.

Response-based schedules (ratio) repeatedly generated higher levels of response than interval schedules. The highest response rates occurred with response-based schedules that were variable (i.e., variable-ratio; like a slot machine or other gambling device).

Applications in the Real World: Shaping

How do animals learn to do anything more complex than pressing a lever? According to Skinner, animals learn complex behaviors through a process called shaping, also known as successive approximations. Shaping involves reinforcing increasingly complex behaviors, gradually leading to the final desired

behavior. The organism's behavior is shaped through a series of steps until it matches the desired response.

Example: To train a rat to push a lever, the steps look like this:

- Rat turns toward the lever → rewarded
- Rat walks toward the lever → rewarded
- Rat touches the lever → rewarded
- Rat presses the lever → rewarded
- Only a full press is now rewarded — earlier steps no longer earn a reward

This step-by-step reinforcement of closer and closer approximations is called successive approximation — the core mechanism of shaping. Skinner recognized that this step-by-step process of successive approximations is not only applicable to animal training but also to complex human learning. This method is commonly used in animal training, such as in circuses, zoos etc.

In therapy or education, instructors may use shaping to help a child develop social skills or to teach a person new daily routines. The process begins with defining the target behavior, then breaking it down into smaller, manageable steps through task analysis.

Shaping has also been used as a therapeutic tool for treating disorders affecting verbal behavior — for example, it has been used to treat selective mutism, a condition where an otherwise normal child refuses to speak in school.

Applications in the Real World: Token Economy

A token economy is a behavior modification system based on the principles of operant conditioning. It involves using tokens as secondary reinforcement to encourage desirable behaviors. Tokens are tangible items like coins, stars, points, or stickers given as rewards for specific behaviors. These tokens can later be exchanged for primary reinforcers, such as tangible rewards or privileges (money, candy etc.).

Specific behaviors identified as desirable and targeted for reinforcement are called target behaviors. Examples include completing homework, following rules, participating in activities, or exhibiting social skills. The system is widely used in various settings, including schools, psychiatric hospitals, and rehabilitation centers.

For example, in a token economy, hospitalized psychiatric patients may receive reinforcing tokens for activities such as serving meals or cleaning floors. In tightly controlled environments, like state hospitals for long-term psychiatric patients, it is feasible to make almost anything a patient wants contingent on the desired behaviors.

Research supports the effectiveness of token economies in increasing behaviors such as social interaction, self-care, and job performance in severely disturbed patients and individuals with intellectual disabilities. They have also been used to decrease aggressive behavior in children and reduce marital discord (Kazdin, 1977).

Lecture 10: Observational learning / Social Learning

Recap: Lecture 9

The previous lecture detailed the mechanisms of Operant Conditioning:

- Skinner's operant conditioning
- Process of operant conditioning: Reinforcement and punishment
- Schedules of reinforcement
- Applications in real world: Shaping

Key Concepts Overview

- Observational learning/Modelling
- Self-Efficacy
- Mirror neurons and observational learning
- Observational learning and prosocial behavior
- Applications in real world: Violence/aggression in media and children

Albert Bandura & Observational Learning

Albert Bandura (December 4, 1925 – July 26, 2021) developed a comprehensive theory of social learning. Bandura's work on observational learning highlights that a significant portion of our learning occurs through observing and imitating the behavior of others, a process termed "modeling" by Bandura. According to Bandura, more of our learning occurs by observational learning as compared to classical or operant conditioning.

The Bobo Doll Experiment

In one of his famous studies conducted with Walters in 1963, children were divided into two groups:

Experimental Group:

Children observed an adult displaying aggressive behavior towards a plastic doll named 'Bobo,' hitting and kicking it while shouting commands like 'Throw him in the air.' In different versions of the study, Bandura used different types of model (such as Live Model (In-person Demonstration), Film Model (Video Demonstration), and Cartoon Model (Symbolic Model)).

Control Group:

Children did not witness such aggressive play.

Results:

Later, both groups were allowed to interact with the doll. The results showed that children in the experimental group (in all conditions) displayed significantly more aggression towards the doll than those in the control group (Bandura & Walters, 1963). This study demonstrated the power of

observational learning, where children imitated the aggressive behavior they had observed in the adult model.

Mirror Neurons, Imitation, and Observational Learning

A mirror neuron is a type of neuron that exhibits activity both when an animal performs a specific action and when it observes another animal performing the same action. These neurons provide a neural basis for imitation and observational learning, enabling organisms to internally “mirror” others’ actions.

The research conducted by Giacomo Rizzolatti and his colleagues in the 1990s led to the discovery of mirror neurons. These neurons “mirror” the observed action, hence their name. Rizzolatti and his team were originally studying the motor cortex of macaque monkeys, specifically focusing on neurons involved in controlling hand and mouth movements. They used microelectrodes to record the activity of individual neurons in the premotor cortex and the inferior parietal lobule of the monkeys' brains.

During the experiments, the researchers noticed something unexpected. They found that some of the recorded neurons not only fired when the monkeys performed specific hand or mouth movements but also fired when the monkeys observed a researcher or another monkey performing the same movements. This phenomenon was initially observed in relation to actions such as reaching and grasping objects (Source: Harvard University Science In The News).

This discovery led Rizzolatti and his colleagues to propose the existence of mirror neurons. Subsequent research in humans has provided evidence for a similar system in the human brain, known as the mirror system.

Functional magnetic resonance imaging (fMRI) studies have demonstrated that certain brain areas in humans, such as the premotor cortex and the inferior parietal lobule, show activity both when individuals perform actions and when they observe others performing the same actions. This suggests the presence of a mirror-like response in humans (Source: Hamilton (2013). Reflecting on the mirror neuron system in autism: a systematic review of current theories, Development and Cognitive Neuroscience, 3, 91-105).

However, due to the inherent challenges in directly studying individual neurons in the human brain, researchers have yet to provide definitive proof of the existence of mirror neurons in humans at the single-cell level. The majority of studies in humans have relied on non-invasive techniques such as fMRI, which provide insights into brain activity at a larger scale but cannot isolate the activity of individual neurons. The term “mirror neurons” often refers to a broader concept of a network of brain regions involved in action understanding and imitation, rather than solely individual cells. The mirror system in humans is believed to involve a distributed network of interconnected brain regions rather than isolated mirror neurons.

The finding of mirror neurons had significant implications for our understanding of human cognition and social interaction. It provided a neural mechanism for how humans can understand and mimic the actions of others, contributing to the development of social skills, language acquisition, and the ability to empathize with others.

Developmental Evidence:

- Infants (8–16 months) imitate new gestures (Jones, 2007, 2017)
- By 12 months, they follow others' gaze (Meltzoff et al., 2009)
- By 14 months, they imitate behaviors seen on TV (Meltzoff & Moore, 1997)

Mirror neuron activity also explains emotional contagion and empathy:

- People unconsciously mimic others' expressions and behaviors
- Yawning spreads across humans and animals (Palagi et al., 2020)
- Observing pain activates similar brain responses as experiencing it (Smith et al., 2021)

Important Factors in Modeling

The modeling process, at first glance, appears simple: an observer and a model, with the primary question being—whether the observer will imitate the model's actions? However, the answer is far from straightforward, as it depends on a multitude of factors. Bandura (1977) concluded from various versions of the study that three factors are crucial in modeling:

(1) Characteristics of the Model:

- Similarity to the Observer: The more similar the model is to the observer, the more likely imitation occurs.
- Complexity of Behavior: Simple behaviors are more likely to be imitated than complex ones.
- Type of Behavior: Hostile and aggressive behaviors are more likely to be modeled.

(2) Attributes of the Observer:

- Confidence Level: Individuals with lower confidence are more prone to imitation.
- Self-esteem: Those with low self-esteem are more likely to imitate.
- Competence: Feelings of incompetence in a situation increase the likelihood of imitation (Low competence → increases dependence on models).
- Learning History: Individuals rewarded for conforming behavior are more inclined to imitate.
- Dependency: Highly dependent individuals are more prone to modeling.

(3) Consequences of Imitating Behavior:

- Positive/negative Results: If individuals believe that imitating a behavior will lead to positive outcomes, they are more likely to do so. According to Bandura, this is the most influential factor.

In a Bobo doll study, some of the children watched as the model who hit the Bobo doll was given praise and a soda and candy. Another group of children saw the model receive verbal and physical punishment for the same aggressive behavior. The children who observed the punishment displayed significantly less aggression toward the Bobo doll than did the children who saw the model being reinforced (Bandura, 1965).

Bandura demonstrates that while reinforcement is essential in classical and operant conditioning, it's not always necessary in observational learning (Bandura, 1977). Attention-grabbing stimuli, like vivid billboard adverts or loud noises, can influence our learning process. Although we may not consciously

process the information at the time, we can recall it later, such as recognizing a product from a billboard when at the supermarket.

Additionally, Bandura shows that individuals can reinforce their own behavior through self-reinforcement (Bandura, 1977). We evaluate our actions, stopping behaviors that provide no pleasure or that we judge as harmful while continuing those that bring positive reinforcement.

The Four Processes of Observational Learning

Bandura analyzed the nature of observational learning and found it to be governed by four related mechanisms (Bandura, 1977): attentional processes, retention processes, production processes, and incentive and motivational processes.

1. Attentional Processes

Attention is the initial stage where the observer focuses on the model's behavior, which is necessary for effective learning to occur. Bandura (1977) identifies various factors that influence attentional processes:

- **Characteristics of the Model:** Observers are more likely to pay attention to models who are similar to them in terms of age, status, sex, or appearance. Celebrity models, experts, and those perceived as confident or attractive often command greater attention.
- **Role of Cognitive and Perceptual Skills:** Observers with highly developed cognitive and perceptual skills and a good understanding of the behavior being modeled are more likely to pay closer attention to the model.
- **Behavior Relevance:** Observers tend to pay greater attention to modeled behavior that they perceive as relevant to their own lives. When the modeled behavior is personally relevant or has implications for the observer, they are more likely to focus their attention on it. Observers pay closer attention to behaviors they expect to perform themselves and those that lead to positive or negative consequences rather than neutral outcomes.

2. Retention Process

The process of retention involves remembering significant aspects of the model's behavior to repeat it later. To retain observed behaviors, the information must be encoded and represented symbolically (Bandura, 1977). This can be achieved through two primary systems: the imaginal internal representational system and the verbal system.

- **Imaginal Internal Representational System:** In this system, individuals form vivid and easily retrievable mental images while observing the model. These mental images allow individuals to summon up a picture of the observed behavior, similar to recalling a memory of a past event (Bandura, 1977). This mental imagery serves as a basis for imitation at a later time. For instance, in observational learning, individuals form a mental picture of the model's behavior, which they can then use as a guide for imitation in the future.
- **Verbal Representational System:** The verbal system involves encoding observed behavior through verbal descriptions or codes (Bandura, 1977). During observation, individuals may describe to

themselves what the model is doing, creating verbal representations of the observed behavior. These verbal descriptions or codes can be rehearsed silently, without overtly displaying the behavior. For example, individuals might mentally rehearse the sequence of behaviors they observed and verbally talk themselves through the steps of a complicated skill. When individuals wish to perform the action later, the verbal codes provide hints, reminders, and cues, facilitating the imitation process.

3. Production Processes

Translating imaginal and verbal symbolic representations into overt behavior requires the production processes, which can be simply described as practice (Bandura, 1977). Despite attending to, retaining, and rehearsing symbolic representations of a model's behavior, individuals may still struggle to perform the behavior correctly, particularly with highly skilled actions that require mastery of many component behaviors. Practice, or production processes, is necessary to translate symbolic representations into actual behavior. This involves physically performing the actions repeatedly to achieve proficiency.

For example, consider the process of learning how to drive a car. While observing a model drive, individuals learn fundamental motions and create symbolic representations of the model's behavior (Bandura, 1977). However, initially translating these symbols into actual driving behavior may be clumsy. For instance, individuals might apply the brakes too soon or too late, or overcorrect the steering. Practice of the proper physical movements, along with feedback on their accuracy, is essential to produce smooth and skilled performance of the behavior.

4. Incentive and Motivational Process

No matter how well individuals attend to and retain observed behaviors or how much ability they have to perform them, they will not do so without incentive or motivation processes (Bandura, 1977). When incentives are available, observation is more quickly translated into action. Incentives also influence attentional and retention processes. Individuals may not pay as much attention without an incentive to do so, resulting in less retention of observed behavior.

- **Anticipation of Reinforcement:** Our incentive to learn is influenced by our anticipation of reinforcement or punishment. Seeing that a model's behavior produces a reward or avoids punishment can strongly incentivize individuals to pay attention to, remember, and correctly perform the same behavior. Reinforcement is experienced vicariously during observation of the model, leading individuals to expect their performance of the same behavior to result in similar consequences.
- **Reinforcement Not Always Necessary:** Bandura pointed out that although reinforcement can facilitate learning, it is not always required. Many factors other than the reward consequences of behavior determine what individuals attend to, retain, and rehearse. For example, loud sounds, bright lights, and exciting videos may capture interest even without immediate reinforcement. Bandura's research demonstrated that children imitate a model's behavior observed on television or in a video game regardless of whether they have been promised a reward. Therefore, while reinforcement can assist in modeling, it is not vital to the process.

Self-Efficacy and Observational Learning

Self-efficacy refers to an individual's belief in their capacity to produce behaviors necessary for specific performance. Albert Bandura has placed the cognitive mechanism of self-efficacy to a central role in his theory of personality functioning and change. While modeling remains significant in his framework, it is primarily viewed as a method to cultivate the necessary level of self-efficacy for effective behavior (Bandura, 1977).

Self-efficacy is a critical component of observational learning. While observational learning explains how we learn a behavior, self-efficacy determines whether we will actually perform that behavior. In Bandura's view, learning is not just a passive imitation of a model. It is a cognitive process where self-efficacy acts as the bridge between "knowing how" and "doing."

The Role of Self-Efficacy in the Four Processes:

- Attention: You are more likely to attend to a model if you believe you have the capability to learn what they are doing.
- Retention: Mental rehearsal of the observed act is more effective when you believe the information is useful and attainable.
- Motor Reproduction: This is where self-efficacy is most visible. Even if you have the "blueprint" of a behavior in your mind, you will not attempt it if you lack the belief in your physical or cognitive ability to execute it.
- Motivation: Self-efficacy provides the internal incentive. If you believe you can succeed (high self-efficacy), you are more motivated to practice and refine the behavior.

Bandura emphasizes that knowing what to do and having the required skills does not guarantee effective performance if individuals have significant self-doubt about their abilities. Consequently, people with the same skills, or the same person at different times, may perform variably—poorly, adequately, or extraordinarily. Competent functioning requires both skills and a sense of self-efficacy to utilize those skills effectively. Additionally, even if people have the necessary skills and a strong sense of efficacy, they may not perform the activities without sufficient incentive (Bandura, 1986, p. 391).

According to Bandura, efficacy expectations shape individuals' choices of activities and environmental settings. These self-efficacy judgments also dictate the amount of effort people will invest in activities and their persistence when encountering challenging tasks and adverse experiences.

People with low efficacy expectations tend to:

- Avoid situations they perceive as threatening and beyond their coping skills.
- When they must engage in such situations, their low self-efficacy leads to minimal effort and quick resignation.

Consequently, they miss out on corrective experiences that could boost their sense of efficacy, remaining defensive and fearful. For example, shy individuals may avoid social settings. If they attend social functions, they exert little effort to interact, often remaining isolated. These individuals are plagued by self-doubt and continuous self-criticism regarding their incompetence, leading to poor

performances in various social contexts. These poor performances reinforce their low self-efficacy and hinder the development of new, assertive behaviors.

People with high efficacy expectations:

- Choose challenging tasks that offer opportunities for skill development.
- In such situations, they exert maximum effort, persist despite obstacles, and engage in minimal self-criticism, increasing their chances of success.

Modeling of Prosocial Behavior

Exposure to prosocial models (helpful, nonviolent behavior) leads to increased helping behavior across cultures and studies (Coyne et al., 2018). Viewing prosocial content in TV, movies, and video games can boost later altruistic actions.

Real-life modeling is equally powerful:

- Observing someone help (e.g., picking up dropped books) increases the likelihood of helping others (Burger et al., 2015; Jung et al., 2020).
- Influential leaders like Mahatma Gandhi and Martin Luther King Jr. used nonviolent modeling to inspire large-scale social change (Matsumoto et al., 2015).

Parents play a critical role:

Individuals who displayed strong moral courage (e.g., rescuers during the Holocaust, civil rights activists) often had parents who modeled humanitarian values (London, 1970; Oliner & Oliner, 1988).

Effectiveness of Models in Observational Learning

Models are most effective when their actions and words are consistent. Children are more likely to adopt behaviors (e.g., religion, persistence) when they see them practiced, not just preached (Lowicki & Zajenkowski, 2020; Butler, 2017). Active participation strengthens learning: for example, parents who practice religious activities with children increase the likelihood of children adopting those beliefs.

Inconsistency (hypocrisy) weakens modeling—but still teaches something: when models say one thing but do another, children often learn both messages (Rice & Grusec, 1975; Rushton, 1975). Children tend to imitate behavior more than words: they may repeat what is said, but more strongly copy what is done. Hence, actions often speak louder than words.

Unfortunately, observational learning can increase antisocial behaviors also, such as abuse, aggression, cheating, lying etc. One such application of observational learning, "The effect of violence shown in media/films/TV on children," is discussed in detail below.

Applications in the Real World: Aggression and Violence in Films, Television, and Video Games

Violence and aggressive acts are prevalent in society, with media playing a significant role in their portrayal. By the time an average child graduates from elementary school, they will have witnessed over 8,000 murders and approximately 100,000 other acts of violence on television (Bushman, 1998, p. 537). This exposure extends to news programs, music videos, law-and-order shows, films, and video games, all of which often depict violent acts.

Critics often argue that media violence has minimal impact on behavior. However, Bandura asserts that media portrayal of violence can have serious and harmful effects. While media violence is not a sufficient or necessary cause of aggression, it can facilitate aggressive behavior in individuals, particularly those who are low in impulse control or high in hypermasculinity (Bandura, 1973, p. 267; Scharrer, 2001, p. 159).

Another common argument is that only disturbed individuals are affected by media violence. Bandura counters this by citing research showing that under certain circumstances, even so-called normal individuals can be induced to act aggressively (Milgram, 1963, pp. 371-378).

Research Evidence:

Bandura's research highlights the profound impact of media violence on behavior, demonstrating that viewers tend to imitate aggressive actions, especially when models are not punished. Observers who see models being rewarded for aggressive behaviors are likely to repeat those behaviors, while those who see models being punished tend not to repeat them. However, even when models are punished, observers can still learn the disapproved behaviors and might repeat them if incentivized (Bandura, 1986, p. 291).

A study found that in more than 5,000 hours of television programming, violence went unpunished 73% of the time (Bushman, 1998, p. 537). Further research revealed that young observers imitated a model rewarded for aggression, even though they reported disliking the model. This suggests that while they understood aggression was wrong, they also found it exciting (Bandura, Ross, & Ross, 1963b, pp. 601-607). This mirrors many television programs where the bad guy is rewarded throughout but punished only at the end, potentially leading viewers to admire the character despite his wrongdoing.

Research since the 1960s and 1970s overwhelmingly supports Bandura's contention that exposure to violent media increases aggression. Studies show that repeated exposure to media violence during childhood correlates with higher aggressiveness in young adulthood and is a significant risk factor for youth violence (Anderson & Bushman, 2001, p. 354).

The late 1970s introduction of violent video games has also been linked to youth aggressiveness. Critics note that shooters in notable school killings, such as those in Paducah, Kentucky; Jonesboro, Arkansas; and Littleton, Colorado, habitually played violent video games. Although the connection is suggestive rather than conclusive, experimental evidence since the mid-1980s indicates that violent video games may indeed increase aggression (Vessey & Lee, 2000, p. 607).

Applications in the Real World: Behavior Modification

Bandura's goal in developing his social-cognitive theory was to modify or change learned behaviors that society deems undesirable or abnormal (Bandura, 1977). Similar to Skinner's approach to therapy, Bandura focused on external aspects, particularly inappropriate or destructive behaviors, operating under the belief that they are learned, just as all behaviors are learned.

Bandura introduced three forms of behavior therapy: modeling, guided participation, and covert modeling. These approaches are effectively applied in addressing fears, phobias, and anxieties, demonstrating Bandura's emphasis on observable behaviors and their modification (Bandura, 1977).

Bandura demonstrated that since modeling is the way we originally learn behaviors, it is also an effective method for relearning or changing behavior. He applied modeling techniques to eliminate fears and other intense emotional reactions.

In a classic study on snake phobia, Bandura and his associates successfully eliminated an intense fear of snakes in adults (Bandura, Blanchard, & Ritter, 1969). The subjects watched a film in which individuals of different age groups made progressively closer contact with a snake, starting from handling plastic snakes to eventually letting a large snake crawl over their bodies. The phobic subjects were allowed to stop the film whenever the scenes became too threatening, and gradually, their fear of snakes was overcome.

The technique called guided participation involves watching a live model and then participating with the model. For example, to treat a snake phobia, subjects watch through an observation window as a live model handles a snake. They then enter the room with the model and observe the handling of the snake at close range. Eventually, wearing gloves, subjects are coaxed into touching the snake while the model holds its head and tail, and they eventually touch the snake without gloves.

Modeling has been shown to be effective even in the absence of an observable model. In covert modeling, subjects are instructed to imagine a model coping with a feared or threatening situation; they do not actually see a model. Covert modeling has been successfully used to treat snake phobias and social inhibitions.

Additional study materials

Feldman, R. S. (2015). *Understanding psychology* (12th ed.). McGraw-Hill Education.

Krosnick, J. A., Betz, A. L., Jussim, L. J., & Lynn, A. R. (1992). Subliminal conditioning of attitudes. *Personality and Social Psychology Bulletin*, 18(2), 152–162.

Other textbooks mentioned in the syllabus